



INTERNATIONAL JOURNAL OF RESEARCH AND STUDIES

Month/Year : June 2026

DOI: <https://doi.org/10.5281/zenodo.21848402>



RECEIVED
20-04-2026



ACCEPTED
23-04-2026



PUBLISHED
30-06-2026

CENTRAL BANK DIGITAL CURRENCY AND THE FUTURE OF BANKING IN INDIA: A CONCEPTUAL PERSPECTIVE

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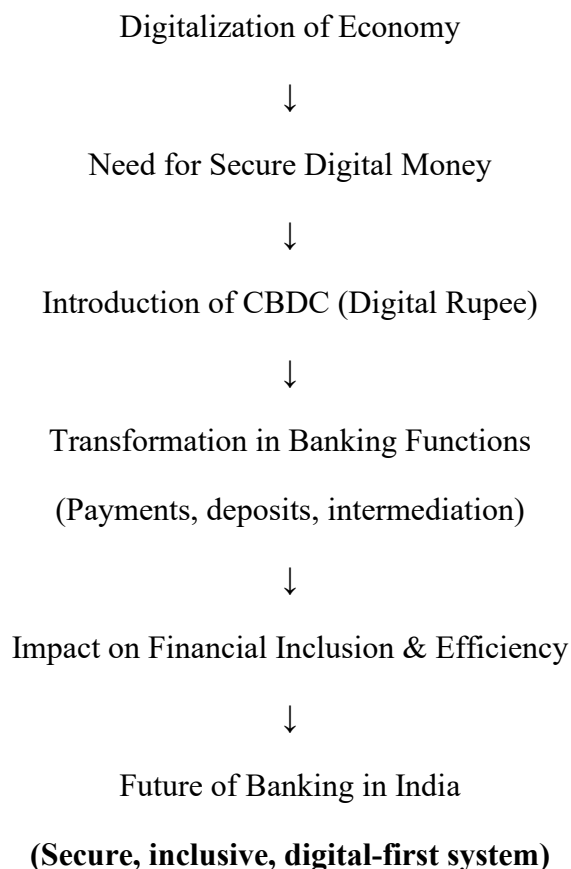
ABSTRACT:

The rapid digitalization of financial systems has accelerated global interest in Central Bank Digital Currency (CBDC) as a sovereign-backed digital form of money. This study examines the concept, features, rationale, and potential implications of CBDC within the Indian banking context. The research adopts a conceptual and descriptive design using qualitative analysis of secondary sources, including RBI reports, international working papers, and academic literature. The study finds that CBDC represents a structural transformation in the nature of sovereign money rather than merely a technological innovation. As a digital liability of the Reserve Bank of India (RBI), the e-Rupee combines legal tender status, central bank backing, programmability, and potential monetary policy advantages. The analysis suggests that CBDC can enhance payment system efficiency by enabling secure, real-time settlement in central bank money while complementing India's advanced digital infrastructure such as UPI and RTGS. In terms of financial inclusion, CBDC has the potential to expand access to risk-free digital money, particularly for underbanked populations, through integration with mobile and digital identity platforms. However, inclusion outcomes depend on digital literacy, infrastructure readiness, and cybersecurity safeguards. The study also highlights the evolving role of commercial banks in a CBDC environment. While concerns regarding deposit disintermediation and digital bank runs exist, a two-tier architecture where the RBI issues CBDC and banks manage distribution can mitigate systemic risks. The findings conclude that CBDC is more likely to transform rather than disrupt India's banking system, provided it is implemented through a phased, well-regulated, and technologically secure framework. Overall, CBDC holds significant potential to strengthen monetary sovereignty, enhance financial efficiency, and support the future of a secure, inclusive, and digital-first banking system in India.

KEYWORDS : Central Bank Digital Currency, e-Rupee, Reserve Bank of India, digital currency, monetary policy, financial inclusion, digital payments, payment efficiency, real-time settlement, Unified Payments Interface, Real Time Gross Settlement, banking transformation, commercial banks.

INTRODUCTION

In, India Central Bank Digital Currency (CBDC) is a legal tender which issued by the Reserve Bank Of India in digital form and its equivalent to physical currency and exchangeable at par (1:1). According to the report that Reserve Bank of India recommended to the center that a person connecting the CBDCs of the BRIC's countries be made part of the agenda for 2026 BRIC's summit in INDIA. This is natural progression of India push during its presidency of the G-20 in 2023 for international cooperation and standardization on cryptocurrency. As digital payments go the UPI infrastructure has proven to be excellent but also for too big a headstart for CBDC overcome. The main difference in CBDC and Cryptocurrencies is that CBDC are issued by a central bank giving consumers guaranteed protection although some concerns remain around data protection and online privacy. Inflation averaged 8.75% globally last year and impacting the quality of life of people all over the world. Where annual jumps of 300% and similarly server drops have left numerous everyday people facing enormous irretrievable losses. the fluctuation of cryptocurrency values is directed mostly by supply and demand for the limited number of coins available. In the context of the banking sector, the Reserve Bank of India (RBI) introduced the CBDC in the year 2022 for there PILOT PROJECT FROM 1st DECEMBER 2022. And section (34) say that the issue department shall be equal to the amount of the currency notes of the government of INDIA and bank note in circulation. and now USA has introduced their CBDC or the digital dollar to its citizens as well as the world. The legal regulators say that the current legislation in some jurisdiction may prevent or the issuance of CBDCs. And the several central bank state the legal foundation to issue CBDC, but the law need to be adjusted in the order to regulate some legal issue which related to CBDC and also specific feature such as programmability. And in the context of CBDC is a global phenomenon and it is very in terms of economies including rapid changes in the payment as well as in the nature of money itself. And there INIDA as set an INTER-MINISTERIAL COMMITTEE, which included the INDIA's central bank and its member briefly examined the implication of CBDC. And the pilot project run across 13 banks in 26 cities which intends to evaluate the digital rupee generations distribution and real – time retail usage process as a whole. and its around 1.46 million customer and 0.31 million merchants are actively in august 31,2023. After India ranked as the third largest fintech worldwide and now INDIA is expecting to make a significant growth to yield \$ 200 billion in profit by 20230. The assumption says that Indian strategy company creating the ground work for startup for private sector fin tech companies to flourish. And by 2030 they collaborate model is expected to generate 13% of the entire revenue generated by fin tech industry worldwide. After the India introduce the online payment systems like UPI, RTGS, NEFT, IMPS, BBPS, and NECT. To shift the digital system has resulted in quick acceptance to provide customer traditional cash and paper transaction. If we see the cyber securities in terms of CBDC it determines that there over 100 central banks around the globe are exploring CBDC to modernize payment system. And CBDC ecosystem is a high value target for a range for a range of a threat action including nation – state and cyber criminals and there is successful attack or operational failure resulting in a service outage, data branch or fraud will erode public trust and confidence with systemic implications. If we go textual conceptual model we



In this flow chart, its determine the purpose of study to know about CBDC. And how digitalization is useful for economy and making future of banking in India, transformation in banking function and the scope of the study is to focus on CBDC in the Indian context and examines conceptual Policy – Level Implication, not empirical performance & its cover the banking payment system & financial inclusion & it does not involve primary data collection.

Review of Literature:

1. Conceptual Foundations of Central Bank Digital Currency (CBDC):

The growing digitization of financial systems has accelerated scholarly discourse on Central Bank Digital Currency (CBDC). CBDC is broadly defined as a digital form of central bank money that is a direct liability of the central bank and denominated in the national unit of account (Auer, Cornelli, & Frost, 2023). Unlike cryptocurrencies, CBDC represents sovereign-backed legal tender and retains the credibility of fiat currency.

Engert and Fung (2017) argue that CBDC emerges primarily in response to declining cash usage, innovations in private digital payments, and concerns about monetary sovereignty. Their work highlights that CBDC is not merely a technological innovation but a structural reform in the monetary system. They emphasize that the introduction of CBDC requires careful evaluation of its implications for payment efficiency, financial intermediation, and central bank balance sheets.

Similarly, Yao (2018) proposes a systematic framework to understand CBDC across four dimensions: currency value, technical design, implementation structure, and application scenarios. He conceptualizes CBDC as credit-based, algorithm-driven, and programmable money, suggesting that its introduction fundamentally redefines the interaction between monetary authority and payment systems. Lee, Yan, and Wang (2021) provide a global comparative perspective, emphasizing the need for a two-tier architecture in which the central bank issues CBDC while commercial banks handle distribution. This layered model ensures stability while leveraging existing banking infrastructure—a design particularly relevant for India’s banking ecosystem.

2. Motivations for CBDC Adoption

The motivations behind CBDC adoption are multidimensional, encompassing payment efficiency, financial inclusion, monetary policy effectiveness, and financial stability. Engert and Fung (2017) identify declining use of physical cash and the rise of private cryptocurrencies as key drivers for central bank intervention. They argue that CBDC can preserve public access to risk-free money in an increasingly digital economy. Williamson (2019) expands the argument by linking CBDC to welfare gains. He contends that digital currency can reduce crime associated with physical cash, enable interest payments on central bank liabilities, and optimize the use of safe collateral in the financial system. These welfare improvements are particularly relevant in emerging markets like India, where digitalization and financial inclusion remain policy priorities.

Davoodalhosseini (2018) analyzes CBDC from a monetary policy perspective and demonstrates that if CBDC bears interest, it enhances central bank policy transmission. His model suggests that CBDC can allow more efficient allocations than cash-based systems, potentially improving welfare outcomes. Auer, Cornelli, and Frost (2023) document global CBDC projects and find that retail CBDC development is more advanced in economies with large informal sectors and high mobile penetration. This observation is significant for India, where digital payments infrastructure (e.g., UPI) coexists with a substantial informal economy.

3. CBDC Design Architecture and Technological Considerations

The design of CBDC significantly influences its economic impact. Zhang and Huang (2022) analyze blockchain-based CBDC models and conclude that permissioned blockchain systems are more appropriate than permissionless ones for central bank implementation. They highlight scalability, performance, and interoperability as critical technical challenges.

Shekhar and Ramesh (2025) provide empirical insights into CBDC design choices with a focus on India’s e-Rupee. Through expert interviews, they propose a four-layer framework encompassing technological infrastructure, security mechanisms, financial integration, and user experience. Their findings emphasize that CBDC should complement rather than replace existing payment systems like UPI.

Lee et al. (2021) stress the importance of interoperability and offline capabilities to ensure mass adoption. They argue that CBDC must be integrated within existing banking frameworks to prevent disruption. In the Indian context, this design discussion becomes crucial as the Reserve Bank of India (RBI) pilots both retail and wholesale e-Rupee models. A two-tier, non-interest-bearing structure aligns with global best practices to prevent excessive disintermediation of commercial banks.

4. CBDC and Financial Stability: Bank Runs and Disintermediation:

One of the most debated aspects of CBDC is its potential impact on banking stability. Critics argue that CBDC may increase the risk of bank runs due to its risk-free nature.

Bitter (2024) develops a dynamic bank-run model and finds that while CBDC reduces bank net worth in normal times, it may actually delay the onset of bank failures during crises, depending on central bank balance sheet adjustments. Her findings challenge the conventional fear that CBDC inherently destabilizes banks. Similarly, Keister and Monnet (2022) identify two countervailing effects:

1. CBDC may reduce maturity transformation, making banks less vulnerable.
2. CBDC provides policymakers with better information flows, enabling earlier intervention.

These findings suggest that CBDC does not automatically increase fragility; instead, design and policy frameworks determine its impact. Elsayed and Nasir (2022) argue that CBDC requires further exploration regarding its implications for financial stability, monetary transmission, and unconventional monetary policy. They emphasize unresolved concerns about privacy, ethics, and technological sustainability. For India, where public confidence in banking remains strong and deposit insurance exists, the risk of digital bank runs may be mitigated through transaction limits and non-interest-bearing CBDC structures.

5. CBDC and Monetary Policy Transmission:

CBDC can transform monetary policy implementation. Davoodalhosseini (2018) shows that an interest-bearing CBDC enhances policy flexibility and allows central banks to achieve first-best allocations under certain conditions. However, he warns that coexistence of cash and CBDC may sometimes reduce welfare. Williamson (2019) argues that CBDC enables direct interest rate application on digital balances, potentially eliminating the zero lower bound constraint. Engert and Fung (2017) caution that widespread CBDC adoption may require central banks to expand their asset holdings to offset deposit displacement from commercial banks.

For India, CBDC could strengthen monetary transmission by reducing friction in digital transactions and enabling programmable payments, particularly in government subsidy transfers.

Research Gap Identified

While global literature extensively covers CBDC design, monetary implications, and financial stability, limited research contextualizes these issues specifically within India's evolving digital ecosystem. Few studies conceptually integrate CBDC with India's existing payment infrastructure, banking structure, and financial inclusion agenda.

Research Objectives

1. To conceptually examine the concept, features, and rationale of Central Bank Digital Currency in the Indian banking context.
2. To analyse the potential implications of Central Bank Digital Currency on the future of banking in India, particularly in terms of payment systems, financial inclusion, and the evolving role of banks.

Research Methodology

The present study, titled "Central Bank Digital Currency and the Future of Banking in India: A Conceptual Perspective," adopts a conceptual and descriptive research design. The purpose of the study is not to test a specific empirical hypothesis or generate primary statistical evidence, but rather to critically examine, interpret, and synthesize existing theoretical frameworks, policy discussions, and institutional perspectives related to Central Bank Digital Currency (CBDC), with special reference to India. A conceptual design is particularly

appropriate in emerging areas of research such as CBDC, where policy evolution, technological experimentation, and regulatory debates are ongoing and where scholarly inquiry is still developing.

The research follows a qualitative and analytical approach. Instead of relying on numerical data or econometric modelling, the study focuses on interpretative examination of ideas, arguments, regulatory frameworks, and institutional structures. The qualitative orientation allows for an in-depth exploration of how CBDC may influence the structure, functioning, and future trajectory of the Indian banking system. Analytical reasoning is employed to evaluate competing viewpoints such as the potential for financial disintermediation versus enhanced monetary transmission and to assess their relevance within the Indian financial ecosystem. The approach facilitates a critical comparison between global CBDC models and India's specific institutional context, including its digital payment infrastructure, banking penetration, regulatory framework, and financial inclusion initiatives.

The study is based entirely on secondary data sources. These include official publications and discussion papers issued by the Reserve Bank of India (RBI), reports and working papers from international institutions such as the Bank for International Settlements (BIS), the International Monetary Fund (IMF), and central banks of other countries, as well as peer-reviewed academic journal articles, policy briefs, and research working papers. Government policy documents, CBDC pilot reports, speeches by policymakers, and scholarly literature on digital currency, monetary policy, financial stability, and banking intermediation are also examined. The selection of sources is guided by relevance, credibility, and scholarly contribution. Preference is given to authoritative institutional documents and peer-reviewed research to ensure conceptual robustness and reliability.

The methodology employed is conceptual analysis combined with systematic synthesis of existing theoretical and policy perspectives. Conceptual analysis involves identifying key constructs such as digital legal tender, two-tier architecture, monetary transmission mechanism, financial disintermediation, bank run dynamics, and programmability of money. These concepts are examined in relation to established theories of banking intermediation, monetary economics, and financial stability. The study then synthesizes diverse strands of literature to build a coherent understanding of how CBDC could reshape the Indian banking landscape. Rather than merely summarizing existing studies, the research integrates them into a structured analytical narrative, highlighting areas of consensus, debate, and research gaps.

A comparative dimension is also incorporated within the conceptual framework. International experiences such as China's digital yuan, Sweden's e-krona project, and other retail and wholesale CBDC initiatives are examined to draw lessons applicable to India. This comparative perspective strengthens the analytical depth of the study and provides contextual grounding for evaluating India's e-Rupee initiative. However, the objective is not to conduct a cross-country empirical comparison but to derive conceptual insights that inform the Indian context.

The study also employs logical deduction and policy interpretation techniques. It critically evaluates how the introduction of CBDC may influence key dimensions of banking in India, including deposit mobilization, credit creation, payment settlement systems, monetary policy transmission, and financial inclusion. By linking theoretical propositions with institutional realities, the research aims to project potential structural transformations in the banking sector.

Since the study relies on documented sources and publicly available information, issues of ethical concern related to human subjects do not arise. Nevertheless, due care has been taken to accurately interpret and

represent the viewpoints expressed in policy and academic documents. Proper citation practices are followed throughout the research to maintain academic integrity.

In summary, the research methodology is designed to provide a comprehensive conceptual understanding of CBDC within the Indian banking framework. By combining qualitative interpretation, analytical reasoning, and synthesis of secondary sources, the study seeks to contribute to policy discourse and academic scholarship on the future of digital currency and banking transformation in India.

Conceptually examine the concept, features, and rationale of Central Bank Digital Currency in the Indian banking context:

Central Bank Digital Currency (CBDC) is broadly defined as a digital representation of sovereign currency issued by the central bank and recognized as legal tender. Unlike cryptocurrencies such as Bitcoin, CBDC is not privately issued nor decentralized; rather, it represents a direct liability of the central bank.

According to the IMF's comprehensive survey on retail CBDC, it is a digital form of central bank money intended for public use and designed to complement or substitute physical cash (Kiff et al., 2020, p. 9). Similarly, Bindseil (2019) describes CBDC as a third form of base money alongside reserves and banknotes, accessible to the general public.

Williamson (2019) conceptualizes CBDC within a multi-means-of-payment framework, where digital currency coexists with physical cash and bank deposits, potentially improving welfare by reducing costs associated with physical currency.

In the Indian banking context, CBDC represents an extension of the RBI's monetary liability into a programmable, digital format. Unlike bank deposits—which are liabilities of commercial banks—CBDC is a claim directly on the RBI, thereby eliminating credit risk. This structural difference is central to understanding its potential impact on banking intermediation.

1. Core Features of CBDC:

Conceptually, CBDC possesses several defining characteristics that differentiate it from both cash and commercial bank deposits.

(i) Legal Tender Status

CBDC is denominated in the national currency and enjoys legal recognition similar to physical cash. It functions as a unit of account, medium of exchange, and store of value (Kiff et al., 2020).

(ii) Direct Central Bank Liability

Unlike bank deposits, which are backed by private banks and subject to default risk, CBDC is backed by the central bank. Fernández-Villaverde et al. (2020) argue that CBDC enables central banks to compete directly for deposits, fundamentally altering financial intermediation structures

(iii) Account-Based or Token-Based Structure

Bindseil (2019) distinguishes between account-based CBDC (centralized ledger accounts) and token-based CBDC (digital bearer instruments). In India, the RBI's design currently follows a token-based retail model while retaining centralized oversight.

(iv) Interest-Bearing Possibility

CBDC can be structured to bear interest. Davoodalhosseini (2018) demonstrates that an interest-bearing CBDC may enhance monetary policy transmission by allowing direct remuneration of digital balances.

(v) Financial Stability Implications

A major theoretical concern is whether CBDC increases bank run risks. Keister and Monnet (2022) show that while CBDC may provide depositors with a safe alternative during crises, appropriate design features can reduce fragility rather than amplify it.

(vi) Impact on Banking Intermediation

Agur, Ari, and Dell'Ariccia (2020) emphasize that CBDC design involves a trade-off between promoting secure public money and preserving bank credit creation. Similarly, Kim and Kwon (2019) note that CBDC may reduce private credit supply unless carefully structured. In the Indian banking system, which relies heavily on deposit mobilization for credit expansion, these features have direct implications for liquidity management, reserve ratios, and maturity transformation.

2. Rationale for CBDC in the Indian Banking Context:

The introduction of CBDC in India is driven by multiple structural and policy considerations.

(i) Strengthening Monetary Sovereignty

With the growth of private cryptocurrencies and stablecoins, central banks risk losing control over monetary transmission. CBDC ensures continued public access to risk-free central bank money (Bindseil, 2019).

(ii) Enhancing Payment Efficiency

India already possesses an advanced digital infrastructure (UPI, IMPS). CBDC can complement these systems by providing final settlement in central bank money, thereby reducing settlement risk.

(iii) Financial Inclusion

Andolfatto (2018) argues that interest-bearing CBDC can enhance financial inclusion by offering government-backed accounts accessible to all citizens. In India, where segments of the population remain underbanked, CBDC could strengthen inclusive access.

(iv) Reducing Costs of Physical Currency

Williamson (2019) highlights that physical currency involves social costs, including theft and illicit use. India's large cash economy implies significant printing, transportation, and storage costs.

(v) Improving Monetary Policy Transmission

Davoodalhosseini (2018) finds that CBDC may enable more effective policy implementation by allowing direct influence over money demand. In India, this could enhance the RBI's ability to manage liquidity and inflation.

(vi) Crisis Management and Stability

Fernández-Villaverde et al. (2020) show that CBDC may deter bank runs due to rigid central bank contracts during panic episodes. For India, which has experienced periodic banking stress, this is particularly relevant.

(vii) International Spillovers

Ferrari, Mehl, and Stracca (2020) argue that CBDC introduction has open-economy implications and may affect international monetary linkages. As India integrates into global financial markets, CBDC may influence cross-border settlements.

3. Conceptual Implications for the Future of Indian Banking:

Conceptually, CBDC does not merely digitize currency—it redefines the relationship between central banks and commercial banks. Three possible structural transformations may emerge:

1. **Complementary Model** – CBDC coexists with bank deposits without significant disintermediation.
2. **Competitive Model** – CBDC competes for deposits, altering banks' funding structures.
3. **Transformative Model** – CBDC reshapes maturity transformation and credit allocation.

The optimal path for India depends on design features such as transaction limits, non-interest-bearing structure, and two-tier distribution through commercial banks.

Potential Implications of Central Bank Digital Currency on the future of banking in India, particularly in terms of payment systems, financial inclusion, and the evolving role of banks:

2.1 CBDC as a New Form of Public Money in the Digital Age:

CBDC is generally conceptualized as a universally accessible form of central bank liability, complementing physical cash and reserves (Kiff et al., 2020). Its integration into payment systems potentially alters the hierarchy of money by enabling direct settlement in central bank money at the retail level.

Bindseil (2019) argues that CBDC may function as a third form of base money, alongside reserves and banknotes, thereby expanding the monetary base accessible to the public. In India, this could lead to: Reduced reliance on intermediary settlement layers, Increased efficiency in final settlement, and enhanced resilience during payment system disruptions.

Ferrari, Mehl, and Stracca (2020) emphasize that CBDC acts as both a payment instrument and a financial asset, combining liquidity and safety features. For India's evolving payment ecosystem, CBDC may provide real-time settlement with reduced counterparty risk.

2.2 Efficiency and Reduced Transaction Costs:

Engert and Fung (2017) identify declining cash usage and payment modernization as major motivations for CBDC. In India, although digital payments are expanding rapidly, cash circulation remains high. CBDC could reduce: Printing and currency management costs, Physical cash handling risks, and informal transaction opacity. Williamson (2019) further argues that CBDC mitigates crime associated with physical currency while

enhancing payment efficiency. Given India's policy emphasis on transparency and digital governance, this feature may support the formalization of the economy.

2.3 Programmability and Smart Payments:

Yao (2018) conceptualizes digital fiat currency as algorithm-based and "smart," enabling programmable applications. In India, this could enable: Conditional government transfers, Direct benefit transfers with spending restrictions, Automated tax compliance mechanisms. Thus, CBDC may extend beyond a payment tool into a policy instrument embedded within digital public infrastructure.

3. Implications for Financial Inclusion:

3.1 Expanding Access to Risk-Free Digital Money

Financial inclusion remains a strategic priority in India. Although initiatives like Jan Dhan Yojana expanded account ownership, usage gaps persist. Andolfatto (2018) demonstrates that interest-bearing CBDC can enhance financial inclusion by providing accessible, government-backed accounts. Similarly, Kiff et al. (2020) emphasize that retail CBDC can broaden public access to central bank money, particularly in underbanked regions. For India, CBDC wallets—if integrated with Aadhaar or mobile infrastructure—could: Provide safe digital savings for unbanked individuals, Reduce dependence on informal financial networks, Strengthen trust in digital payments.

3.2 Narrow vs Broad CBDC and Inclusion Trade-Off

Chen and Siklos (2022) distinguish between a narrow CBDC (replacing cash) and a broader deposit-like CBDC. A narrow CBDC may support financial inclusion without drastically altering banking intermediation, whereas a broad CBDC may compete directly with bank deposits. In India, a calibrated "narrow retail CBDC" may achieve inclusion objectives while minimizing banking disruption.

4. Evolving Role of Banks in the CBDC Era:

The introduction of CBDC raises fundamental questions about the future structure of banking.

4.1 Risk of Disintermediation

Bindseil (2019) highlights the risk of structural disintermediation if CBDC significantly substitutes bank deposits. Similarly, Agur, Ari, and Dell'Ariccia (2020) show that CBDC design must balance financial intermediation with payment diversity. In India, commercial banks rely heavily on deposits to fund credit. Large-scale migration to CBDC could: Reduce deposit bases, Increase funding costs, Constrain lending capacity.

4.2 Stability and Bank Runs

Keister and Monnet (2022) present a nuanced view, arguing that CBDC has two opposing effects: it may increase run incentives, but also allows better monitoring and early intervention. Kim and Kwon (2019) find that CBDC deposits could reduce private credit supply unless central bank lending mechanisms offset liquidity shifts.

Fernández-Villaverde et al. (2020) show that CBDC may stabilize the system during panic episodes due to rigid central bank contracts. Thus, the impact on Indian banks depends critically on design choices such as: Holding limits, Non-interest-bearing structure, Two-tier architecture.

4.3 Two-Tier Model and Role Transformation

Lee, Yan, and Wang (2021) advocate a two-tier or multi-tier architecture, where the central bank issues CBDC but commercial banks handle distribution and customer interface. This model implies that Indian banks may evolve into: Service providers and wallet managers, KYC and compliance intermediaries, Liquidity distributors rather than exclusive deposit holders. Hence, banks may shift from balance-sheet dominance to platform-based intermediation.

5. Broader Macroeconomic and Policy Implications

Ferrari et al. (2020) highlight that CBDC may amplify monetary policy transmission and international spillovers. Davoodalhosseini (2018) further shows that interest-bearing CBDC strengthens policy control. For India, this implies; Improved liquidity management, enhanced inflation targeting efficiency, and greater digital traceability of money flows. However, Viñuela, Sapena, and Wandosell (2020) caution about the “CBDC dilemma”—balancing financial stability and disintermediation risks.

Finding, Conclusion, and Suggestions:

The study finds that Central Bank Digital Currency (CBDC) represents a structural evolution in the nature of sovereign money rather than merely a technological innovation. In the Indian context, CBDC (e-Rupee) is conceptualized as a digital form of legal tender issued by the Reserve Bank of India (RBI), functioning as a direct liability of the central bank and equivalent to physical currency. The analysis reveals that CBDC possesses distinct features—legal tender status, central bank backing, programmability, and potential interest-bearing capability—which differentiate it from both commercial bank deposits and private cryptocurrencies.

The findings indicate that the introduction of CBDC has significant implications for India’s payment ecosystem. CBDC has the potential to enhance settlement efficiency, reduce transaction costs, minimize reliance on physical cash, and strengthen the resilience of digital payment infrastructure. Given India’s strong digital foundation through systems such as UPI, IMPS, RTGS, and NEFT, CBDC is more likely to function as a complementary instrument rather than a substitute for existing platforms. Furthermore, programmability features could improve targeted government transfers, subsidy distribution, and transparency in public finance.

In terms of financial inclusion, the study finds that CBDC may expand access to safe and risk-free digital money, particularly for underbanked and digitally connected populations. Integration with mobile infrastructure and digital identity systems can strengthen inclusive finance. However, inclusion benefits depend on digital literacy, cybersecurity preparedness, and accessibility in rural areas.

Regarding the evolving role of banks, the study identifies a potential structural shift in banking intermediation. CBDC may lead to partial deposit substitution if not carefully designed, thereby affecting banks’ funding capacity and credit creation functions. However, the two-tier model—where RBI issues CBDC and commercial banks manage distribution—appears to mitigate the risk of disintermediation. The research suggests that banks may gradually transition from traditional deposit-based intermediaries to technology-enabled financial service providers focusing on customer interface, compliance, and credit intermediation.

The study also finds that CBDC may strengthen monetary policy transmission by enabling more direct liquidity management and improving transparency in money flows. However, financial stability concerns such as digital bank runs, cybersecurity threats, and privacy risks require careful regulatory safeguards.

Overall, the findings suggest that CBDC in India is more likely to transform rather than disrupt the banking system, provided it is introduced through a calibrated and well-regulated framework.

Conclusion:

The introduction of Central Bank Digital Currency marks a significant milestone in the evolution of India's monetary and banking architecture. This study concludes that CBDC is not simply a digital replacement for cash but a foundational innovation that redefines the interaction between central banks, commercial banks, and the public. In the Indian context, the e-Rupee reflects a strategic move toward strengthening monetary sovereignty, enhancing payment system efficiency, promoting financial inclusion, and supporting digital governance.

The conceptual analysis demonstrates that CBDC can coexist with existing digital payment systems and may serve as a secure settlement layer within India's digital financial ecosystem. While concerns about financial disintermediation and bank runs are theoretically valid, these risks can be managed through design features such as transaction limits, non-interest-bearing structures, and a two-tier operational model.

The future of banking in India under a CBDC framework is likely to be characterized by digital-first operations, platform-based intermediation, enhanced regulatory oversight, and improved policy transmission mechanisms. Rather than eliminating commercial banks, CBDC may reshape their functions toward greater efficiency, innovation, and customer-centric services.

Thus, the study concludes that CBDC has the potential to strengthen India's banking system, provided its implementation is gradual, well-regulated, technologically secure, and aligned with broader financial stability objectives.

Based on the findings and conclusions of the study, the following suggestions are proposed: First, the RBI should continue adopting a phased and calibrated approach to CBDC expansion. Gradual scaling of pilot projects will allow policymakers to assess operational risks, cybersecurity vulnerabilities, and behavioral responses before full-scale implementation. Second, maintaining a two-tier architecture is strongly recommended. Commercial banks should remain central to CBDC distribution, customer management, and liquidity intermediation to prevent structural disintermediation and preserve credit creation capacity. Third, transaction limits and non-interest-bearing features should be maintained in the early stages to minimize deposit migration from banks and reduce the risk of digital bank runs during periods of financial stress. Fourth, a strong cybersecurity infrastructure must be prioritized. Since CBDC systems are high-value targets for cyber threats, investment in encryption, resilience testing, and real-time monitoring mechanisms is essential to maintain public trust. Fifth, financial literacy and digital awareness programs should be strengthened, especially in rural and semi-urban areas. Without adequate digital literacy, the financial inclusion benefits of CBDC may remain limited. Sixth, regulatory clarity regarding data privacy, consumer protection, and cross-border usage must be established to address legal uncertainties and enhance confidence in the system. Finally, continuous research and empirical evaluation should accompany CBDC deployment. Future studies should examine behavioral adoption patterns, macroeconomic impacts, and long-term implications for banking stability and monetary policy effectiveness in India.

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